

Nikhil Mandayam Adyapak

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SUMMARY

Master's student in Data Science at UW-Madison with 2+ years of experience as a Senior Software Engineer building scalable MLOps platforms and distributed AI systems. Specialized in the engineering side of AI, including architecting cloud-native training pipelines on Azure Kubernetes Service (AKS) and optimizing GPU infrastructure. Proven track record of reducing training cycles and engineering efficient data retrieval systems for ADAS.

Seeking 2027 Full-Time Roles in ML, ML Systems, Software, MLOps or Data Science. Available June 2027.

EDUCATION

University of Wisconsin-Madison

Sep 2025 - May 2027

Masters in Data Science

GPA - 3.68/4.00

Coursework: Machine Learning, Foundation Models, Database Design & Implementation, Statistical Models, Statistical Methods, Statistical Inference for Data Science

PES University - Bengaluru, India

Aug 2019 - May 2023

Bachelor of Technology in Computer Science & Engineering

GPA - 3.78/4.00

Specialization: Machine Intelligence & Data Science

Coursework: Algorithms, Data Structures, Data Science, Linear Algebra, Machine Learning, Big Data, Cloud Computing

TECHNICAL SKILLS

Programming languages: Python, C, C++, Java, SQL, R

ML & Statistics: PyTorch, TensorFlow, HuggingFace, Transformers, LLM, CNN, GAN, scikit-learn, OpenCV, matplotlib

MLOps & Workflows: Ray, MLFlow, DVC, Argo Workflows, Azure ML Studio, Prometheus, Grafana

Cloud: Azure, AWS, AKS, Docker, Terraform, Azure DevOps, GitHub Actions, Linux

Databases & Tools: ElasticSearch, PostgreSQL, MySQL, MongoDB, FastAPI, Flask, Streamlit, GitHub

RESEARCH EXPERIENCE

Bosch Global Software Technologies (BGSW)

Bengaluru, India

Senior Software Engineer – AI Systems for Advance Driver Assisted Systems (ADAS)

Aug 2023 – Aug 2025

- Engineered **scene-understanding** image retrieval pipeline using foundation models (**Mask2Former**, **Depth-Anything**, **OWL-ViT**, **CLIP**) to generate scene graphs, curating fine-grained datasets for ADAS corner cases.
- Benchmarked **Occlusion Estimation** models on the **nuScenes** dataset by engineering custom LiDAR-depth evaluation pipelines to generate metrics and visualizations.
- Enhanced ADAS image dataset resolution by **4x** utilizing **ESRGAN** transfer learning on curated datasets.
- Researched optimal deployment strategies for multi-tenant **Ray Clusters** on **on-premise workstations** and **Azure Kubernetes Service (AKS)** to efficiently support **ML Workloads**.
- Integrated **CLIP embedding** re-use with **80%** cross-version alignment, avoiding full re-indexing of legacy image vectors.

Center for Information Security Forensics and Cyber Resilience, PES University

Bengaluru, India

Summer Research Intern; Advisor: Prof. Vineetha B, Dr. Prasad B Honnavalli

May 2021 – Nov 2021

- Developed a novel cryptanalysis algorithm in **Python** to decipher single columnar transposition ciphers without prior knowledge of the encryption key or its length.
- Engineered an optimization approach reducing computational overhead by dynamically determining key lengths via matrix permutations and longest-prefix dictionary matching.

INDUSTRY EXPERIENCE

Ciroos

Pleasanton, CA (SF Bay Area)

AI Intern

May 2026 – Dec 2026

- Productionized a **Graph RAG** context engine on a temporal knowledge-graph (**Graphiti+Neo4j**) sourcing SRE agents incident answers in team chat and investigation history (**94%** accuracy).
- Prototyped a graph-free **Postgres RAG** (LLM distillation, **hybrid vector + full-text retrieval**, RRF fusion, LLM reranking), **84%** accuracy over incident chats at **27x** lower ingestion cost.
- Conducted performance evaluations (**evals**) of AI-driven Site Reliability Engineering (SRE) agents by simulating faults across Kubernetes and multi-cloud environments (**AWS, GCP, Azure**).
- Authored and fine-tuned detailed instruction pipelines that guide AI agents to accurately perform Root Cause Analysis (**RCA**) and fault diagnosis across varied system domains.

Bosch Global Software Technologies (BGSW)

Bengaluru, India

Senior Software Engineer – AI Systems for Advance Driver Assisted Systems (ADAS)

Aug 2023 – Aug 2025

- Delivered **MLOps platform** for pedestrian detection (**Azure ML Studio + MLFlow**) with end-to-end **ML lifecycle**; enabled **multi-GPU training** (**3x** faster) and cut GPU costs by **60%**.

- Designed and scaled a **Ray Cluster** based distributed **ML training** system on **Azure Kubernetes Service (AKS)**, reducing retraining cycles from **4 weeks** to **1 week (75% faster)** and supporting **30+** teams.
- Architected a containerized **FastAPI** backend for a **60M+ vector image-search engine**, deploying **CLIP embeddings** and **ElasticSearch** to reduce edge-case triage by **95%** (4 hours to 10 mins).
- Automated **ML pipeline** CI/CD workflows on **AKS (Terraform, Helm, Argo, GitHub Actions)**; integrated **Prometheus** alerting, accelerating experiment setup for **multi-tenant** ADAS teams from **3 days** to **5 minutes**.
- Deployed on-prem **Ollama LLM** with Python FastAPI wrapper, enabling private, team-wide LLM access (**150+ users**), zero external data egress.

Machine Learning Operations (MLOps) Intern

Jan 2023 – May 2023

- Built on-premise **MLOps** pipelines (**DVC, MLflow**) and accelerated **multi-GPU** object detection training (YOLOv5, Detectron2) on **Azure ML Studio**, cutting setup time from **1 week** to **1 day (85% faster)** and training time by **67%**.
- Containerized models (**Docker**) and engineered CI/CD workflows via **Azure DevOps** for automated deployment to **Azure Container Registry (ACR)**.
- Presented the MLOps proof-of-concept at the internal Data Engineering Marketplace, demonstrating production readiness and driving team adoption.

Summer Intern

Jun 2022 – Jul 2022

- Designed an initial MLOps pipeline architecture for object detection and segmentation (Detectron2, YOLOv5) with DVC.
- Implemented ensemble strategies, performance visualizations, and automated evaluation reports to improve model robustness for autonomous driving scenarios.

PROJECTS AND PUBLICATIONS

[1] **Code Runtime Complexity Prediction** - ERCICA (Springer), 2023 | Python, TensorFlow, sklearn, Streamlit, NetworkX
Predicted Big-O runtime complexity of C/Java/Python code using static analysis and ML classification with BiLSTM over Abstract Syntax Trees graph embeddings on IBM CodeNet dataset (**Accuracy: 96%**). [\[View Publication\]](#)

[2] **Sentinel-Edge AI** - MadData Hackathon 2026 (Qualcomm Track) | Llama 3.2, Langchain, RAG, FastAPI
Selected as 1 of 10 teams to build an air-gapped **Edge AI** security auditor on Snapdragon X Elite NPU; engineered a RAG pipeline with ChromaDB to detect legal risks and code vulnerabilities without cloud egress. [\[View Project\]](#)

[3] **MLOps POC pipeline for Pedestrian Detection** - 2023 | Python, DVC, MLFlow, DAG, Streamlit, SQLite, PyTorch
Implemented on-premise MLOps starter template with DVC + Detectron2 for end-to-end ML lifecycle. [\[View Project\]](#)

[4] **Novel ways of decrypting transposition ciphers**, - IEEE smartgencon, 2022 | Python, Optimization, Natural Language
Introduced cipher-breaking optimization techniques over columnar transposition ciphers. [\[View Publication\]](#)

TEACHING EXPERIENCE

Teaching Assistant, CS564: Database Management Systems

Madison, WI, USA

University of Wisconsin-Madison; Professor: Xiangyao Yu

Jan 2026 – May 2026

Lead assignment sections, mentor students, and hold office hours to support database projects and assignments. (**SQL, Python**). Design and grade assignments, exams, and project rubrics; provide feedback to **200+** students to improve understanding of database design and query optimization. Assist course instructors with course logistics, and automated grading pipelines.

Teaching Assistant, CS203: Statistics for Data Science

Bengaluru, India

PES University; Professors: Dr. Bharathi R, Swati M

September 2022 – Dec 2022

Responsibilities included teaching, designing, and grading coursework, assignments, and projects. Delivered hands-on sessions on **Data Science topics** such as **hypothesis testing, statistics, and supervised ML**. Organized, managed, and served as a panelist for a Hackathon. Mentored **120+** students enrolled in the undergraduate level Data Science course.

PRESENTATIONS AND COMPETITIONS

Wisconsin Experience Bus Tour (WEBT) – Leadership Trip

Wisconsin, USA

University of Wisconsin-Madison, Student Leadership & Involvement (SOLI)

May 2026

- Selected for the **fully-sponsored** 5-day statewide leadership trip (35 students) to engage with industry, cultural, and community leaders across Wisconsin.

MadData Hackathon 2026 (Qualcomm Track)

Madison, WI, USA

University of Wisconsin-Madison & Qualcomm

Feb 2026

- Selected as **1 of 10 teams** to build privacy-first Edge AI applications on the **Snapdragon X Elite NPU**.
- Developed "Sentinel-Edge AI," an air-gapped security auditor using local **Llama 3.2, Langchain** and **RAG** to analyze sensitive legal/code documents with zero cloud data egress.

3 Minute Thesis (3MT) Competition

Madison, WI, USA

University of Wisconsin-Madison

Nov 2025

- Selected to present research on "Generating Fine-Grained Datasets for Autonomous Driving" using vision foundation models.
- Condensed technical research into a compelling 3-minute narrative for a general, non-technical audience.

PROFESSIONAL AWARDS AND SCHOLARSHIPS

Top Achiever (1st/150+) & Best Presenter Award	Bosch Data Engineering Dept.	2024
Distinction Award (Academic Excellence, Sem 3-6)	PES University	2022-2023
MRD Scholarship (40% Tuition, Top 20% of 1100+)	PES University	2020

VOLUNTEERING EXPERIENCE

CSR Volunteer	Bengaluru, India
Bosch Global Software Technologies (BGSW)	Oct 2023 – Mar 2025
• Tech Mentorship & Education: Mentored candidates with disabilities in tech interview preparation (Youth4Jobs), taught Computer Science to underprivileged students, and developed interactive materials for rural childcare centers. • Social Impact: Supported agricultural initiatives funding stipends for differently-abled individuals and conducted educational surveys at government boys’ homes. • Community & Wildlife: Assisted in elephant habitat restoration, volunteered for abandoned dog adoption drives (CUPA), and led infrastructure renovation and sports coaching at local government schools.	